additional figures

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This document contains the code to reproduce the plots in the manuscript that do not directly present *results*, i.e. figures 1, 2, 3, and 8. The code to reproduce figures 4 through 7 that present actual results of the key empirical analyses are in separate documents alongside the code to reproduce those results ("method_evaluation" and "phylogenetic_examples").

Note that two R packages are required: `EloSteepness` (https://cran.r-project.org/package=EloSteepness) and `EloSteepness.data` (https://github.com/gobbios/EloSteepness.data).

```
library(EloSteepness)
library(EloSteepness.data)
```

figure 1

```
# generate example ratings
r <- c(770, 875, 1100, 1400)
rats <- round((r - mean(r)) / 200, 2)
names(rats) <- LETTERS[4:1]

layout(matrix(c(1, 1, 2, 3, 4, 4), ncol = 2, byrow = TRUE), heights = c(1, 5, 1))

# ratings
par(mar = c(0, 4.5, 0, 4.5), family = "serif", cex = 1.2)
plot (0, 0, "n", xlim = c(0, 5), ylim = c(0.5, 2.5), axes = FALSE, ann = FALSE)
rect(-2, 0.5, 7, 2.5, border = NA, col = grey(0.9), xpd = TRUE)
text(1:4, rep(1, 4), rats)
text(1:4, rep(2, 4), LETTERS[4:1], font = 2)
text(-0.4, 1.5, label = "Elo-rating", adj = 0.5, xpd = TRUE)

# winning probabilities matrix
par(family = "serif", mar = c(3, 4, 3, 2), cex = 1)
mat <- round(outer(rev(rats), rev(rats), function(x, y)plogis(y - x)), 2)
# mat <- mat[4:1, 4:1]
mat <- t(mat)
colnames(mat) <- rownames(mat) <- LETTERS[1:4]

twocols <- hcl.colors(4, "zissou1")[3:4]

n <- ncol(mat)
colmat <- mat
colmat[, ] <- "black"
colmat[3, ] <- twocols[1]
colmat[1, ] <- twocols[2]
diag(colmat) <- NA
diag(mat) <- ""
plot(0, 0, "n", xlim = c(0.5, n + 0.5), ylim = c(n + 0.5, 0.5), ann = FALSE, axes = FALSE)
xmat <- sapply(seq_len(n), function(x) rep(x, n))
ymat <- t(xmat)
text(xmat, ymat, mat, col = colmat)
cn <- colnames(mat)
text(rep(0.1, n), seq_len(n), cn, xpd = TRUE, font = 2, col = colmat[, 2])
text(seq_len(n), rep(0.4, n), cn, xpd = TRUE, font = 2)
title(ylab = "winning probability", xpd = TRUE, line = 1.4)

# winning probabilities, sigmoid
par(family = "serif", mar = c(4, 3, 1, 1), mgp = c(1.5, 0.5, 0), tcl = -0.2)

xseq <- seq(-6, 6, by = 0.2)
plot(xseq, plogis(xseq), type = "n", ylim = c(0, 1), xlim = c(-6, 6), xaxs = "i",
     yaxs = "i", las = 1, xlab = "rating difference", ylab = "winning probability")
points(xseq, plogis(xseq), type = "l", xpd = TRUE, col = "black", lwd = 3, lend = 3)

id <- 2
xvals <- rats[id] - rats[-id]
yvals <- plogis(xvals)
```

```
arrows(xvals, rep(0, 3), xvals, yvals, code = 2, length = 0.1, xpd = TRUE, col = twocols[1])
arrows(xvals, yvals, rep(-6, 3), yvals, code = 2, length = 0.1, xpd = TRUE, col =  twocols[1])

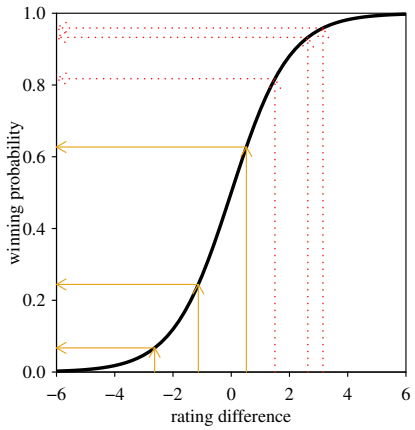
id <- 4
xvals <- rats[id] - rats[-id]
yvals <- plogis(xvals)
arrows(xvals, rep(0, 3), xvals, yvals, code = 2, length = 0.1,
       xpd = TRUE, col = twocols[2], lty = 3, lwd = 1.5)
arrows(xvals, yvals, rep(-6, 3), yvals, code = 2, length = 0.1,
       xpd = TRUE, col = twocols[2], lty = 3, lwd = 1.5)

# summed/cumulative winning probs
par(mar = c(0, 4.5, 0, 4.5), cex = 1.2)
plot (0, 0, "n", xlim = c(0, 5), ylim = c(0.5, 2.5), axes = FALSE, ann = FALSE)
rect(-2, 0.5, 7, 2.5, border = NA, col = grey(0.9), xpd = TRUE)
text(1:4, rep(1, 4), rev(colSums(round(outer(rev(rats), rev(rats),
                                             function(x, y)plogis(y - x)), 2)) - 0.5))
text(1:4, rep(2, 4), LETTERS[4:1], font = 2)

text(-0.4, 1.5, label = "summed winning\nprobability", adj = 0.5, xpd = TRUE)
```

Elo−rating	D	C	B	A
	−1.33	−0.81	0.32	1.82

winning probability	A	B	C	D
A		0.82	0.93	0.96
B	0.18		0.76	0.84
C	0.07	0.24		0.63
D	0.04	0.16	0.37	



summed winning probability	D	C	B	A
	0.57	0.94	1.78	2.71

figure 2

```
set.seed(123)
cols <- sample(hcl.colors(6, "zissou1", alpha = 0.7))

data("toy_example", package = "EloSteepness.data")

matlist <- list(toy_example$m1, toy_example$m2)
reslist <- list(toy_example$e1, toy_example$e2)

par(mfrow = c(2, 3), family = "serif")

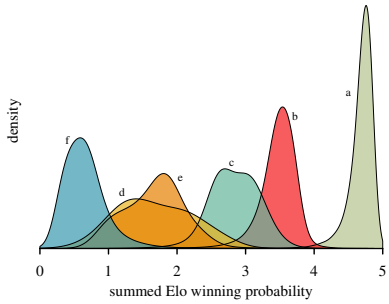
par(mar = c(3, 3, 4, 1))
EloSteepness:::plot_matrix(matlist[[1]], greyout = 0)
mtext("winner", 2, line = 1.5, cex = 0.8)
mtext("loser", 3, line = 1.5, cex = 0.8)
par(mar = c(4, 3, 1, 1))
EloSteepness::plot_scores(reslist[[1]], color = cols, adjustpar = 4)
text(4.5, 2, "a", cex = 0.8)
text(3.72, 1.7, "b", cex = 0.8)
text(2.8, 1.1, "c", cex = 0.8)
text(2.05, 0.9, "e", cex = 0.8)
text(1.2, 0.73, "d", cex = 0.8)
text(0.4, 1.4, "f", cex = 0.8)

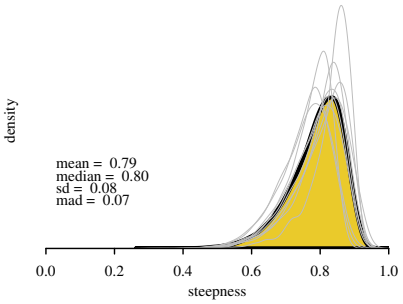
EloSteepness::plot_steepness(reslist[[1]])

par(mar = c(3, 3, 4, 1))
EloSteepness:::plot_matrix(matlist[[2]], greyout = 0)
mtext("winner", 2, line = 1.5, cex = 0.8)
mtext("loser", 3, line = 1.5, cex = 0.8)
par(mar = c(4, 3, 1, 1))
EloSteepness::plot_scores(reslist[[2]], color = cols, adjustpar = 4)
text(4.7, 5.2, "a", cex = 0.8)
text(3.98, 1.8, "b", cex = 0.8)
text(2.9, 1.7, "c", cex = 0.8)
text(1.95, 0.95, "e", cex = 0.8)
text(1.1, 0.85, "d", cex = 0.8)
text(0.5, 3, "f", cex = 0.8)

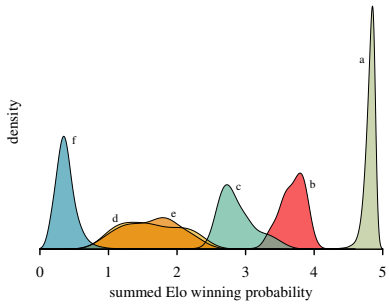
EloSteepness::plot_steepness(reslist[[2]])
```

winner \ loser	a	b	c	d	e	f
a		2	3	4	1	2
b	0		1	1	1	1
c	0	0		1	1	1
d	0	0	0		1	1
e	0	0	0	1		1
f	0	0	0	0	0	





winner \ loser	a	b	c	d	e	f
a		4	6	8	2	4
b	0		2	2	2	2
c	0	0		2	2	2
d	0	0	0		2	2
e	0	0	0	2		2
f	0	0	0	0	0	



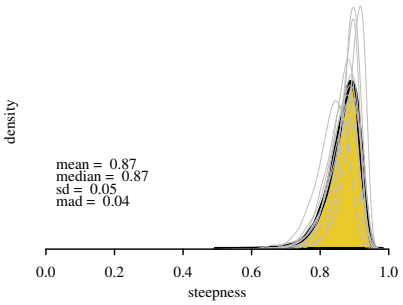


figure 3

```
set.seed(123)
cols <- sample(hcl.colors(6, "zissou1", alpha = 0.7))

par(mfrow = c(2, 2), family = "serif", mgp = c(1.5, 0.5, 0), tcl = -0.2)
data("toy_example", package = "EloSteepness.data")

EloSteepness::plot_steepness_regression(x = toy_example$e1, color = cols)
text(1.4, 4.8, "a", cex = 0.8)
text(2.3, 3.9, "b", cex = 0.8)
text(3.2, 3.3, "c", cex = 0.8)
text(4.3, 2.4, "e", cex = 0.8)
text(4.7, 2.6, "d", cex = 0.8)
text(6.1, 1.0, "f", cex = 0.8)

d <- EloRating::DS(toy_example$m1, prop = "Pij")
plot(0, 0, type = "n", xlim = c(1, 6), ylim = c(0, 5), las = 1, axes = FALSE,
     xlab = "ordinal rank", ylab = "normalized David's score", yaxs = "i")
axis(1, at = 1:6)
axis(2, at = 0:5, las = 1)
box(bty = "l")

r <- lm(d$normDS ~ I(rank(d$normDS * (-1))))
abline(r)
points(rank(d$normDS * (-1)), d$normDS + c(0, 0, 0, 0.1, -0.1, 0),
       pch = 21, bg = "white", col = adjustcolor(cols, alpha.f = 1), lwd = 3, cex = 1.5, xpd = TRUE)
text(1, 4.5, "a", cex = 0.8)
text(2, 3.5, "b", cex = 0.8)
text(3, 2.5, "c", cex = 0.8)
text(4.4, 1, "e", cex = 0.8)
text(4.6, 2, "d", cex = 0.8)
text(6, 0.5, "f", cex = 0.8)



EloSteepness::plot_steepness_regression(x = toy_example$e2, color = cols)
text(1.4, 4.65, "a", cex = 0.8)
text(2.3, 4.0, "b", cex = 0.8)
text(3.1, 3.3, "c", cex = 0.8)
text(4.3, 2.3, "e", cex = 0.8)
text(4.7, 2.5, "d", cex = 0.8)
text(6.1, 0.9, "f", cex = 0.8)

d <- EloRating::DS(toy_example$m2, prop = "Pij")
plot(0, 0, type = "n", xlim = c(1, 6), ylim = c(0, 5), las = 1, axes = FALSE,
     xlab = "ordinal rank", ylab = "normalized David's score", yaxs = "i")
axis(1, at = 1:6)
axis(2, at = 0:5, las = 1)
box(bty = "l")

r <- lm(d$normDS ~ I(rank(d$normDS * (-1))))
abline(r)
```

```
points(rank(d$normDS * (-1)), d$normDS + c(0, 0, 0, 0.1, -0.1, 0),
       pch = 21, bg = "white", col = adjustcolor(cols, alpha.f = 1), lwd = 3, cex = 1.5, xpd = TRUE)
text(1, 4.5, "a", cex = 0.8)
text(2, 3.5, "b", cex = 0.8)
text(3, 2.5, "c", cex = 0.8)
text(4.4, 1, "e", cex = 0.8)
text(4.6, 2, "d", cex = 0.8)
text(6, 0.5, "f", cex = 0.8)
```

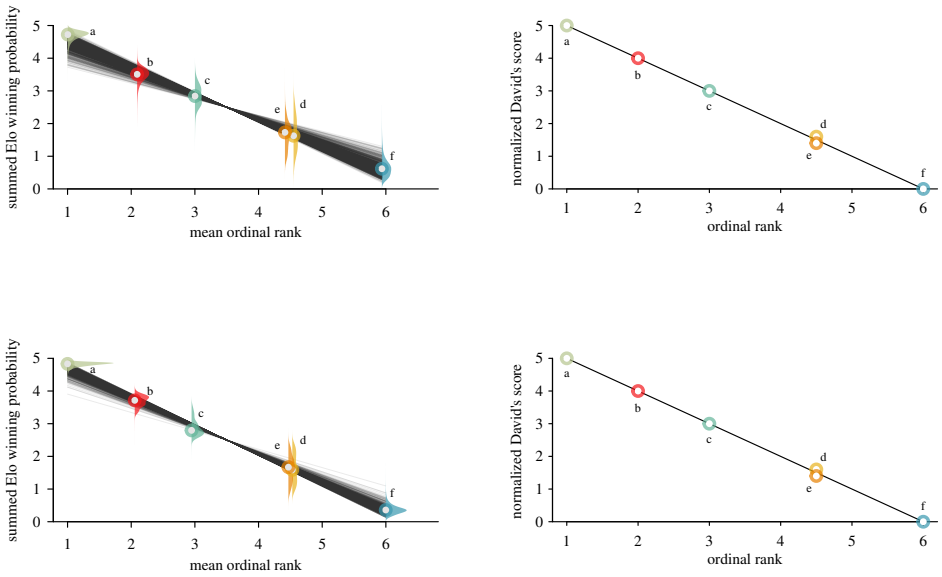


figure 8

```
par(mfrow = c(1, 2), family = "serif", mgp = c(2, 0.7, 0), tcl = -0.3)

data("cred_width", package = "EloSteepness.data")
# limit table to initial matrices
pdata <- cred_width[grepl("_step00", cred_width$mat), ]

plot(log(pdata$i_ratio), pdata$median, xlab = "log interactions/group size ('data density')",
     ylab = "Elo steepness (median of posterior)",
     las = 1, lwd = 0.4, xlim = c(0, 7), xaxs = "i",
     ylim = c(0, 1), bty = "l", yaxs = "i", xpd = TRUE, cex = 0.7)
abline(v=log(c(10, 20)), lty = c(1, 2), col = "grey")

plot(log(pdata$i_ratio), pdata$cred_width89, xlab = "log interactions/group size ('data density')",
     ylab = "Elo steepness (width of 89% credible interval)",
     las = 1, lwd = 0.4, xlim = c(0, 7), xaxs = "i",
     ylim = c(0, 1), bty = "l", yaxs = "i", xpd = TRUE, cex = 0.7)
abline(v=log(c(10, 20)), lty = c(1, 2), col = "grey")
```

